

QUIZ 04 PRACTICE PROBLEMS – Solutions

STAT 1110: Section S18 – Fall 2026 with Dr. Ethan P. Marzban

⚠ Caution

These problems are not meant to be exactly indicative of what will show up on the Quiz on Monday, but are rather designed to be a starting point for your studies.

Use the following setup for Questions 1 through 6:

Shivani wants to determine whether the average length (in minutes) of a *Queen*¹ song is less than 5 minutes or not. She has collected some data, and used the Theory-Based Inference applet to obtain the following results, with some inputs redacted:

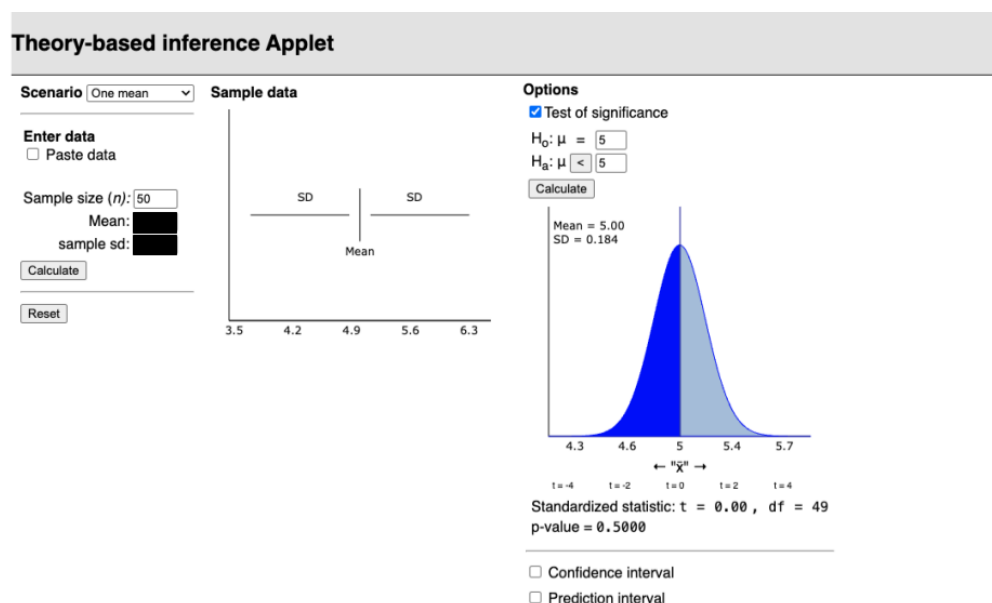


Figure 1: A screenshot from the Theory-Based Inference Applet.

If it is difficult to read: the sample size is indicated to be 50; the hypotheses have a null value of 5, and the sign in the alternative is $>$; the standardized statistic is listed as $t = 0.00$; the p -value is listed as 0.5000; the mean and SD of the null distribution are specified to be 5.00 and 0.184, respectively.

1. Which of the following best describes an observational unit in the context of this dataset?

- (A) One song
- (B) A collection of 50 songs
- (C) The length of a song
- (D) The average length of a collection of 50 songs

¹*Queen* is a popular musical band from the 80s and 90s

Solution: One observational unit is one song.

2. What is the variable, and is the variable quantitative or categorical?

(A) The variable is the average length (in minutes) of a collection of 50 songs; quantitative

(B) The variable is the length (in minutes) of a song; quantitative

(C) The variable is whether or not a given song is less than 5 minutes in length; categorical

Solution: A variable is just an attribute/characteristic recorded from/on each observational unit. Each observational unit, in the context of this study, is one song; from each trial we only record the length (in minutes) of that particular song. This is a **quantitative** variable since values are numbers, and it makes interpretive sense to perform arithmetic operations on the values.

3. What is the value of $\bar{x}$?

(A) 0.000

(B) 0.184

(C) 5.000

(D) 50.00

Solution: Because the t -statistic value is 0.00, this means $\bar{x}$ is equal to the null value, which in this case is **5**.

4. What is the value of s , rounded to two decimal places?

(A) 0.00

(B) 0.19

(C) 1.30

(D) 5.00

Solution: We know that

$$SD_{\text{NULL}} = \frac{s}{\sqrt{n}}$$

Rearranging terms gives

$$s = \sqrt{n} \times SD_{\text{NULL}}$$

which, plugging in relevant values, gives

$$s = \sqrt{50} \times (0.184) \approx 1.30$$

5. Consider the following four statements:

I) Because the p -value is relatively large, there is very strong evidence against the null

II) Because the p -value is relatively large, there is weak evidence against the null

III) Because the t -statistic is relatively small in magnitude, there is very strong evidence against the null

IV) Because the t -statistic is relatively small in magnitude, there is weak evidence against the null

Which of the following correctly quantifies the strength of evidence, for the correct reasons? Select only ONE of the options below.

- (A) (I) and (III)
- (B) (II) and (IV)**
- (C) (I) only
- (D) (IV) only
- (E) (III) only

Solution: Large p -values provide weak evidence against the null; t -statistics with small magnitude also provide weak evidence against the null.

6. Which of the following best describes a Type I error in the context of this study?

- (A) Shivani rejects the null.
- (B) Shivani rejects the null when the null was actually true.
- (C) Shivani concludes that the average length of a Queen song is less than 5 minutes, when, in fact, the true average length of a Queen song is 5 minutes.**
- (D) Shivani concludes that the average length of a Queen song is equal to 5 minutes, when, in fact, the true average length of a Queen song is less than 5 minutes.

Solution: Though Choice (B) is technically correct, it is not written **in the context of this study** (which is something we expect you to do when phrasing your own conclusions!)